AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2019

Computer Science Paper II

Time: 2 hours 15 minutes Marks: 55

April/ May 2019 for Learning only

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are TEN questions. Answer ALL questions. Question 10 offers TWO choices. Attempt any ONE choice.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid, or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 5 Marks)

a. Cathode Ray Tube (CRT) monitors are being replaced by Liquid Crystal Display (LCD) monitors in homes and offices.

Write any THREE reasons to justify the given statement. (3 Marks)

_____

_____

_____

_____

_____

_____

b. Windows is the most commonly used operating system around the world. However, some other operating systems also exist.

List the names of any TWO operating systems other than Windows operating system. (2 Marks)

_____

_____

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Q.2. (Total 5 Marks)

a. Identify whether each of the given characteristics in the table is of a dot matrix printer or a laser printer. (2 Marks)

S. No.	Characteristic	Dot Matrix Printer/ Laser Printer
1.	It uses papers with small holes on each side to feed them in printer.	
2.	It is non-impact printer.	
3.	It prints an entire page at a time.	
4.	It is used when low quality prints are required at slow print rate.	

b. Show whether the given statements are true or false with the help of working.

(**Note:** No marks will be awarded if working is not shown.) (3 Marks)

Statement	True/ False	Working
4 gigabytes capacity of Flash Drive is enough to store 130 sound files of size 50 megabytes each.		
A link having data rate of 8 megabits per second will transfer a video file of 10 megabytes in 10 seconds.		
A portable hard disk has 1.5 terabytes of capacity which is equal to 1500000 megabytes.		

Q.3. (Total 5 Marks)

Describe any FIVE advantages of setting up computer network in an office.

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Q.4. (Total 5 Marks)

Differentiate between Local Area Networks (LANs) and Wide Area Networks (WANs) on the basis of the features in the given table.

Feature	LAN	WAN
Speed		
Security		
Setup Cost		
Coverage Area		
Transmission Errors		

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Q.5. (Total 6 Marks)

Define the following terms with reference to data transmission.

- Bandwidth
- Modulation
- Guided Media
- Baseband Signal
- Serial Transmission
- Synchronous Transmission

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Q.6. (Total 5 Marks)

Computer aided design (CAD) is a software that is used to create drawings and designs that are used by computer aided manufacturing (CAM) to control the machine tools.

With reference to the given statement, describe any FIVE features that must be present in CAD software.

Q.7. (Total 5 Marks)

A teacher wants to use a word processor to simplify his/ her routine tasks in the school.

Describe any FIVE distinct features of word processor that will help the teacher to achieve this.

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Q.8. (Total 4 Marks)

Ahmed is working on MS Word file that has two text boxes, **A** and **B**. He wants to perform a number of tasks using keyboard only.

Write a keyboard shortcut that Ahmed should apply to

a. make the text in the text box **A** same as text in the text box **B.** (1 Mark)

I love Pakistan	*I Love Pakistan*
A	**B**

_____

_____

b. change the font size of text from 15 to 11 in the text box **B** of part (a). (1 Mark)

_____

_____

c. underline the text in the text box **B** of part (a). (1 Mark)

_____

_____

d. retrieve the accidently deleted text from the text box **B**. (1 Mark)

_____

_____

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Q.9. (Total 5 Marks)

	A	B	C	D	E
1	**Training Participants Data**				
2	**Name of Participant**	**Name of School**	**Gender**	**Age**	**Qualification**
3	Mubashir Ahmed	Bright Light School	M	32	MA
4	Rana Asad	New Horizon School	M	26	MA
5	Fatima Khan	Pure Pakistani School	F	29	BA
6	Shumaila Ali	The Best School	F	30	MA
7	Aqeel Ahmed	New Horizon School	M	28	BA
8	Amna Siddiqui	Bright Light School	F	27	MA
9	Javed Ahmed	The Best School	M	31	MA

Consider the given Excel sheet containing Training Participants' Data and write MS Excel function with appropriate cell ranges and values to calculate the

a. total number of participants. (1 Mark)

b. average age of participants. (1 Mark)

c. age of the youngest participant. (1 Mark)

d. number of participants whose qualification is MA. (1 Mark)

e. number of participants with age more than and equal to 30. (1 Mark)

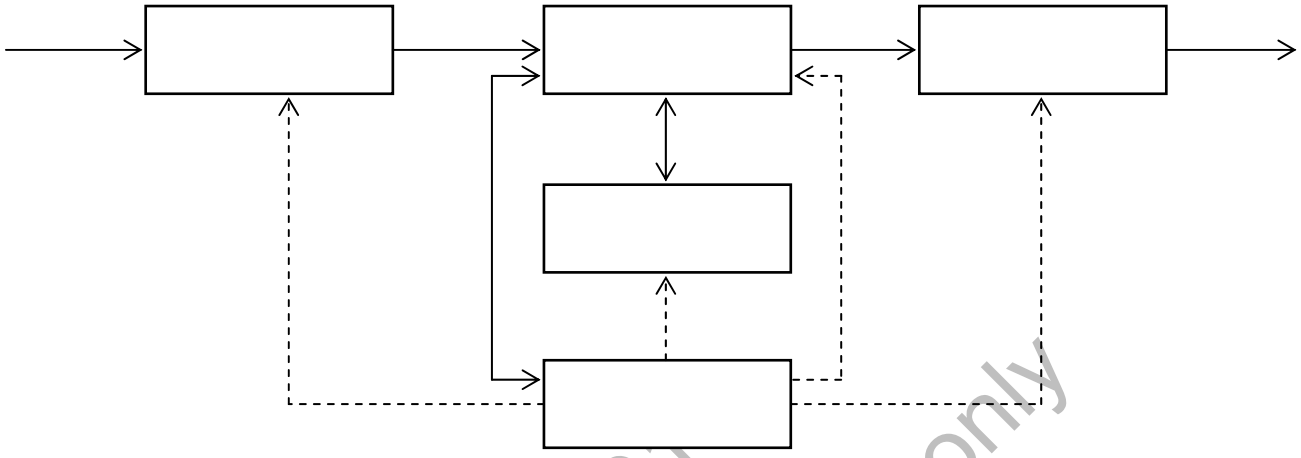
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Q.10. (Total 10 Marks)

EITHER

a.

i. Consider the given blank block diagram of a computer system. (5 Marks)



(**Note:** In the given diagram, the data line is represented by ⟺ and the control line is represented by ----> .)

Read the terms listed below and complete the above blank block diagram by placing the **number** of each term in appropriate box.

S. No.	Term
1.	Control unit
2.	Memory unit
3.	Arithmetic logic unit
4.	Register
5.	Input unit
6.	Output unit

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ii. Describe any TWO differences between random access memory (RAM) and read only memory (ROM). (2 Marks)

iii. Describe the purpose of each of the given buses in a computer system. (3 Marks)

- Data bus
- Address bus
- Control bus

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OR

b.

i. Describe any SIX different computer crimes. (6 Marks)

ii. List any FOUR data protection rules to keep data safe. (4 Marks)

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END OF PAPER

Please use this page for rough work

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